Receipted Recursive Recovery of Negative Quantum-Method Results

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Abstract

Negative results in computational research are often recorded only as prose, discarded after a method changes, or reopened without an explicit rule for what changed. We study an executable alternative in this work: prospectively freeze the question, comparison set, gates, and honest terminals; give the strongest admissible donor method first right of refusal; retain nulls, refutations, and donor absorptions as first-class records; type the failure and authority state that permits reopening; and bind result-bearing computations to replayable receipts. The contribution is methodological rather than a new quantum algorithm. Across one end-to-end programme, this discipline produces all of the outcomes it is supposed to permit. Several plausible R6 candidate families are exactly absorbed or refuted before a bounded structural residual is isolated; the N1–N4 recovery studies contain both donor closures and exact-synthetic positives; protected data remain sealed when a frozen gate does not license access; and capability or orchestration success never self-grants scientific authority. We use these cases to separate three properties that are often conflated: a result can be reproducible without being scientifically valid, scientifically negative without being an execution failure, and methodologically informative without being a novel method. The evidence is a one-programme case study, not a statistical evaluation of research-methodology effectiveness. Claims owned by the companion TARE, dual-instrument, and typed-state papers are treated only as outcomes generated under the common discipline. A fresh literature closure remains a submission gate for any novelty statement about the methodology itself.

Keywords: negative results; research provenance; reproducibility; autonomous research; prospective protocols; donor comparison.

1 Introduction

A negative result has at least three possible meanings: the candidate mechanism is wrong, the current representation/search family is inadequate, or the computation could not establish either conclusion. Research software commonly collapses these states. A failed run may be treated as negative evidence; a new idea may silently replace the comparison set; a donor-equivalent method may be described as new; or an old negative may be reopened without recording which assumption changed.

Companion work studies a narrower object than a general theory of scientific discovery: an executable discipline for preserving and recursively revisiting negative computational results. The discipline is built around five constraints. Outcome rules are frozen before result access. Donor methods receive first right of refusal. Exact claims carry witnesses or independently checkable referees. Failure and authority are typed so a capability result cannot self-authorize a scientific conclusion. Finally, the record is replayable enough that a later reader can reconstruct what was known, what was attempted, and why a lane stopped.

The this programme programme provides a stress test because its history contains the outcomes that a non-gamed method should retain: donor absorptions, exact refutations, bounded positives, mixed results, and unresolved/cannot-check states. The strongest mathematical result of that programme is reported separately; the dual-instrument benchmark is reported separately; the partial-knowledge mechanism studies belong to separate work. Here they matter only as evidence that the common methodology did not force them into one preferred terminal.

The central claim is therefore descriptive and methodological: this receipt-first negative-recovery workflow was instantiated end-to-end and behaved in a non-monotone, outcome-sensitive way under prospectively frozen rules. We do not estimate how often such a workflow improves science, and we do not claim that receipt completeness is a substitute for evidentiary quality.

2 Methods

2.1 Prospective freeze

Each result-bearing lane binds its subject, admissible information, candidates/baselines, endpoints, gates, tie-breaks, and terminal vocabulary before outcome inspection. Later corrections are logged rather than retroactively changing a gate. This makes a refutation or donor tie an admissible result rather than an incentive to move the goalposts.

2.2 Donor first right of refusal

A candidate receives scientific residual credit only after the strongest faithfully composable existing mechanism has been run or otherwise discharged under the same admitted information. A tie or donor win closes the candidate claim. Donor absorption is therefore a positive methodological terminal even though it yields zero candidate novelty credit.

2.3 Typed failure and authority

The workflow separates scientific state from orchestration state. Missing capabilities, failed tools, and malformed receipts are control conditions, not evidence against a hypothesis. Scientific/candidate receipts cannot by themselves grant novelty, adoption, promotion, merge, or global-stop authority. Reopening is tied to typed changes in assumptions, interfaces, representations, or evidence rather than to generic dissatisfaction with a negative result.

2.4 Exactness and hostile controls

Where a claim is finite and exact, the programme uses exhaustive enumeration, proof-carrying dynamic programming, independent witness reconstruction, or complete local machine checks. Protocols predeclare hostile controls designed to make an attractive shortcut fail. If the control does not bite where required, the world/lane is invalid rather than counted as a positive.

2.5 Receipt and replay contract

Result-bearing computations serialize canonical inputs, outputs, identifiers, and authority boundaries. Content-addressed or hash-bound receipts permit deterministic replay or independent recomputation. The contract establishes attribution and reproducibility; it does not establish the truth of a scientific inference without the corresponding design/referee.

2.6 Case-study selection

We analyze the complete R6 saturation/recovery arc and the N1–N4 closure families because they were defined by the programme before this paper synthesis and include heterogeneous terminals. The paper does not select only successful recoveries. Companion papers own the domain-specific claims.

3 Results: behavior of the methodology

3.1 R6: repeated refusal before a bounded residual

The R6 chain first produces donor absorption and several exact negatives: progressively richer joint compiler families add no value beyond stronger donor constructions on the frozen open subjects. A support-dominance audit then converts repeated collapse into a structural explanation while explicitly leaving a Tag-coupling gap. The first closure claim fails at that declared gap; its repair is later refuted by a converse coupling; support-two enlargement restores exactness on the tested domains and is subsequently upgraded to an all-n theorem in the companion separate work line.

For separate work the important observation is procedural. The workflow reports the early candidate families as donor-equivalent or refuted rather than promoting them as separate methods, yet it does not stop searching merely because several negatives accumulated. Each next question is licensed by a specific residual left by the preceding result.

3.2 N1–N4: recovery is not a one-directional success story

The N-lane studies exercise the same discipline on different partial-knowledge and synthesis questions. N1 contains a bounded typed-failure-state benefit whose allocation policy is exactly closed by an ideal value-of-information donor. N2 includes a prospectively supported crossover predictor that is subsequently donor-absorbed on the original world, leaving only a narrower misspecification result. N3's registered synthesis families survive their donor comparisons on the stated exact-synthetic domains. N4's six partial-knowledge families support typed/scoped state mechanisms against matched-information controls while retaining designed negative regimes.

These outcomes cannot be pooled into a meaningful single success fraction: they test different questions, and donor absorption is intentionally a scientifically different terminal from candidate support. Their common methodological value is that the record preserves the distinction.

3.3 Protected and orchestration boundaries

The R6 prospective gate records an honest 'R6_EARNED = NO' outcome while the protected stretched-N2 discriminator remains sealed. The harness/campaign records also keep host or capability failures outside the scientific evidence channel. This is evidence that the implementation can refuse to turn unavailable execution or unreleased protected data into a favorable scientific terminal.

3.4 Replayability

The publication packet indexes protocols, result receipts, and replay checks across the case studies. Deterministic lanes reproduce canonical outputs; exact claims include independent or alternate reconstruction where specified. Replay discrepancies, if any, are a publication blocker rather than something to be averaged away.

4 Discussion

The case studies suggest that recursive recovery is most useful when it is treated as a rule-governed change of question, not as persistence. Reopening after a donor closure without a new residual would be search inflation. Refusing to reopen after a representation or access boundary changes would treat an old negative as globally permanent. Typed state and prospectively declared reopen conditions provide the bridge between those extremes.

The R6 history illustrates a second distinction: saturation and explanation are not the same. Several exact negatives can establish that a family adds no value on a scope, but a later dominance theorem or counterexample is needed to explain why. A receipts-first archive makes that transition possible because the negative sequence remains available to be compared with the later structural account.

Third, the method separates provenance from authority. Reproducible evidence can still be badly designed, use a weak donor, or support only a narrow inference. Conversely, an honest ‘CANNOT_CHECK‘ can be methodologically correct even though it offers no scientific answer. A reviewer should therefore evaluate the scientific design of each companion result independently of separate work's provenance claim.

The unresolved research question is comparative: does this discipline improve the reliability, efficiency, or calibration of research programmes relative to other workflows? The current evidence does not answer it. Doing so would require a prospectively defined multi-programme evaluation with matched tasks and independently adjudicated outcomes.

5 Related work and claim boundary

The workflow combines established research practices rather than claiming them individually. Preregistration formalizes the separation between hypotheses/analyses defined before outcomes and post-outcome explanations [1]. Scientific-computing guidance emphasizes version control, testing, automation, and recoverable workflows [2]; data-provenance research formalizes how derived outputs remain traceable to source records [3]. Tool-using autonomous scientific systems have also demonstrated that language-model agents can plan and execute multi-step research activities [4]. These literatures motivate components of this work but do not establish the particular recursive negative-recovery contract studied here.

The candidate residual is their executable composition around recursive negative-result recovery: a typed terminal system in which donor absorption, refutation, support, and cannot-check remain distinct; reopening is tied to a recorded residual; and the evidence path is receipt-bound. That residual is conditional on a fresh submission-date literature search. The manuscript therefore avoids ‘first’, ‘unique’, or exhaustive-nearest-work language until that search is complete.

A companion paper establishes the exact TARE expressivity result, and another reports the controller–host agreement benchmark and is bounded away from adjacent systems-paper territory.

Separate work covers the typed/scoped partial-knowledge mechanism evidence. This paper cites those outcomes only to show the behaviour of the common methodology.

Failure records as a shared asset, and what is still missing. The closest existing system to the one studied here is Wang's negative-knowledge memory layer for automated research [5]. A curator agent, deliberately separate from the agent that ran the attempt, converts each failed attempt into a bounded record in a shared bank; the failure is classified against a closed vocabulary over layer, scope, degree, recommended action and risk, so the curator must commit to a failure class rather than describe it in prose; and a downstream agent must explicitly adopt or reject each relevant record, with justification, before proposing its next experiment. The reported effect is real: on a scientific-coding benchmark, structured records raise retry pass rate at lower token cost than raw self-debug, and a bank built on one coupled PDE system transfers to a mechanistically different one.

That work establishes most of what this paper would otherwise have to argue from scratch — that failures are worth typing, that typing them beats prose, that a separate curator reduces self-assessment bias, and that the resulting records are reusable rather than merely archival. It is treated here as the strongest parent, not as a competing claim.

Three things it does not settle, and which this paper is about. First, the records are written *after* an attempt finishes, so nothing constrains what counted as failure before the outcome was seen; the discipline studied here freezes the question, the comparison set, the gates and the honest terminals prospectively. Second, a record classifies what failed but carries no authority state governing when the route may be reopened, so a bank can be revisited at any time for any reason; here the reopening condition is itself typed, and a terminal cannot be reopened without an explicit change in the authority that closed it. Third, the records summarise artifacts rather than binding to them, so a record and the computation it describes can drift apart; here result-bearing computations are bound to replay receipts. The difference is between a memory layer that makes failures reusable and a provenance discipline that makes them checkable, and the second is what the recursive-recovery claim requires.

An institutional remedy for the same problem. The gap addressed here has been named at the level of the publication system. Schaeffer et al. argue that machine-learning conferences accept flawed work and then provide no mechanism to correct the record: there are no editorial boards able to attach errata, nothing compels withdrawal, and critiques are therefore displaced to blogs and social media, where they lack archival permanence and expert scrutiny [6]. Their proposed remedy is a dedicated refutations track. Two of their observations matter here independently of that proposal. First, they separate refutation from reproducibility: a result can be reproducible and technically correct while the surrounding claim is misleading, so re-running an experiment is not the same as testing what was asserted about it. Second, they note that a refutation is only meaningful against a claim stated precisely enough to be refuted — a paper that does not define what it asserts cannot be shown wrong.

Both observations are prerequisites for the discipline studied here, and neither is supplied by it. The difference is where the correction lives. A refutations track is a venue: it creates a place for a human critique to be reviewed and cited, and its guarantees are social ones about visibility and standing. What is studied here is machine-checkable and local to a project: the terminal that records a negative result is typed, the authority permitting the route to reopen is typed alongside it, and the computation behind the record is bound to a replay receipt. That does not replace an institutional mechanism, and it does not confer the visibility that publication does. It addresses the part their proposal explicitly leaves to authors, which is whether the record inside a project is in a state where a refutation could be checked at all.

6 Limitations

The evidence comes from one programme in one repository, with many protocols authored by the same research effort that later synthesizes the methodology. This supports implementation and case-study claims, not an unbiased estimate of effectiveness. The case studies are heterogeneous and cannot be converted into an aggregate recovery rate without defining a population and sampling process that do not currently exist.

Prospective freezing reduces outcome-contingent gate changes but does not guarantee that a frozen question is well chosen. Donor first refusal is only as strong as the donor search and faithful implementation. Replay guarantees attribution/reconstruction but not construct validity, external validity, or novelty. Typed authority prevents one class of self-promotion but does not replace independent scientific review.

Several studies use exact-synthetic worlds, and some live-host actions depend on model/tool behavior that is replayed from receipts rather than re-executed identically. The methodology should therefore be tested on independently designed programmes before any broader reliability claim.

7 Conclusion

this programme demonstrates an executable way to preserve and recursively revisit negative computational research results without forcing every lane toward a positive. The method freezes outcome rules, gives donors first refusal, retains negative history, types the state that licenses reopening, and binds computations to replayable records. In the observed programme it yields donor absorptions, exact refutations, bounded positives, protected-data restraint, and honest unresolved states.

That is the paper's claim ceiling. The work does not establish that this workflow is generally superior to alternative research methodologies, nor that a receipted result is automatically good evidence. Its value is a reproducible contract for saying what was tried, what failed, why a question was reopened, and which stronger claim remains unauthorized.

8 Reproducibility

The paper-level `RECEIPT_INDEX.md` is the publication map from claims to protocols/results. Before submission it must be regenerated or manually reconciled against the final manuscript so every headline statement resolves to a committed receipt and every receipt cited by a headline statement has an independent replay or reconstruction route.

The R6 chain lives under `development/orion-q-max-r0/` and `research/extensions/orion-q/`; the N-lane protocols/results live under `development/orion-q-nlane-closure/` and `research/extensions/orion-q/nlanes/`. Harness-level replay artifacts are under the corresponding development directories. The root `REPRODUCE.md` gives the required verification classes and failure semantics.

Reproduction is claim-sensitive. A byte-identical receipt validates replay; an exact scientific claim additionally requires its referee/witness contract; a donor claim requires the donor comparison; protected-custody claims require evidence that the protected reference remained unreleased.

9 Ethics, safety and resource statement

The studies reported here do not involve human participants, animals, clinical data, or personal data. The relevant integrity risks are selective reporting, hidden post-outcome gate changes, accidental protected-data access, and authority laundering from a successful tool or model call into a scientific conclusion. The methodology is designed specifically to make these events visible or fail closed.

Computational cost varies by lane from inexpensive deterministic synthetic studies to large exact enumerations. Resource use is not itself evidence. Negative and null computations are retained even when they are expensive, because deleting them would distort the recovery history. No capability receipt, model output, or campaign terminal can by itself authorize novelty, publication, deployment, or scientific truth.

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